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Probability density function of MDOF structural systems under non-normal delta-correlated inputs

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Abstract

A method to approximate the probability density function of MDOF linear systems under non-normal delta-correlated input is presented. The method requires: (i) the evaluation of the response cumulants up a given order, by solving the set of cumulant differential equations; (ii) the evaluation of the quasi-moments of the response by means of recursive relationships, once the response cumulants are known; (iii) the evaluation of the coefficients of the C-type Gram–Charlier series expansion of the response probability density function, by solving a set of linear algebraic equations, whose known terms depend on the quasi-moments of the response up to a given order. The numerical application shows the versatility and the accuracy of the proposed method.

1. Introduction

In many cases of engineering interest it has become quite common to use Gaussian stochastic processes in order to model loading. However, in many real situations the loading process has strong non-Gaussian characteristics. Therefore, the Gaussian hypothesis of the input is not admissible. This occurs when the random excitation is adequately represented by a train of pulses with random amplitudes occurring at random times [1,2]. Examples of engineering problems for which the excitations can be modelled as explained above are: moving loads on a bridge due to highway traffic flow [3,4], buffeting as an intermittent excitation acting on an airplane tail, wind and waves forces acting on structural systems [5,6], etc.

If random occurrences of pulses are assumed to be independent, the occurrence times are described by the Poisson law and the loading process is a non-normal delta-correlated process. The problem of linear dynamic systems under delta-correlated input was dealt with by several authors [7–12]. The probabilistic characterisation of the non-Gaussian response process is carried out in terms of statistical moments or cumulants up to an order greater than two, whose differential equations can be derived by making use of the generalised Itô differential rule [10,13]. Moreover, in order to solve the problems of first passage or threshold crossing, the response probability density function is required; such a function is generally approximated by means of the A-type Gram–Charlier series expansion [1,7,14–16], starting from the knowledge of the quasi-moments up to a given order. The quasi-moments of a given order are related to the statistical moments or cumulants up to the same order. It is known that the above mentioned expansion reveals some serious shortcomings; indeed, regions of negative probability with non-physical meaning may occur; furthermore, should the response process be strongly non-Gaussian, many terms are required to obtain acceptable accuracy.

In this paper the response probability density function is approximated by means of the C-type Gram–

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Charlier series expansion [17]; this expression is more rapidly convergent and ensures a good accuracy of the probability density, which has positive values over the whole range [16,18]. However, as the relationships between the series expansion coefficients and the statistical moments (or the cumulants or the quasi-moments) are not known so far, the use of this expansion has not been extensively applied in stochastic analysis. In this paper it is shown that these coefficients can be evaluated as a function of the quasi-moments of the response, by solving a set of linear algebraic equations, for both the one-dimensional and the multi-dimensional systems. Moreover, the great accuracy of the C-type Gram–Charlier expansion with respect to the traditional A-type expansion is shown in the numerical example.

2. Cumulant differential equations for non-normal delta-correlated input

2.1. One-dimensional linear systems

Let us consider a one-dimensional system governed by the following linear differential equation:

$$\dot{Z}(t) = D(t)Z(t) + V(t)W(t) \tag{1}$$

where $W(t)$ is a non-normal delta-correlated process; this means that the value $W(t_j)$, for any particular time t_j, is statistically independent of the values $W(t_k)$ at any other time or times. This implies that the rth order correlation function of $W(t)$ has the following expression:

$$R_r[W(t_1), W(t_2), \dots, W(t_r)] = q_r(t_1)\delta(t_2 - t_1)\delta(t_3 - t_1)\cdots\delta(t_r - t_1) \tag{2}$$

$\delta(t)$ being the Dirac's delta function and $q_r(t)$ the rth order intensity coefficient. If $q_r(t) = 0$ for $r > 2$, then $W(t)$ becomes a Gaussian white noise process.

A remarkable example of delta-correlated process is represented by the Poisson white noise process, defined as [2,19]:

$$W(t) = \sum_{k=1}^{N(t)} P_k \delta(t - t_k) \tag{3}$$

where $N(t)$ is the counting Poisson process, $\{P_K\}$ is a family of random variables, which are independent of each other and of the time instants t_k; the time occurrences t_k are distributed according to the Poisson law, with arrival rate equal to $\lambda(t)$; in the case of Poisson white noise, the intensity coefficient $q_r(t)$ of order r is simply given by

$$q_r(t) = \lambda(t)E[P^r] \tag{4}$$

where $E[\cdot]$ means stochastic average. If $q_r(t)$ is time independent for every r, then $W(t)$ is a strongly stationary process. In the particular case in which $\lambda(t)E[P^2] = \text{const}$ as λ tends to infinity and $\lambda(t)E[P^r]$ for $r > 2$ vanish, then $W(t)$ approaches to a Gaussian white noise process.

It is well known that the fundamental tool to obtain the differential equations governing the response cumulants evolution of a system under white noise excitations is represented by the Itô differential rule [20]; recently, the Itô differential rule has been modified in order to consider the non-normal delta-correlated excitations [21]. In this way, it is easy to obtain the differential equation governing the evolution of the rth order response cumulant as follows [22]:

$$\dot{k}_r[Z(t)] = rD(t)k_r[Z(t)] + V^r(t)q_r(t) \tag{5}$$

where $k_r[Z(t)]$ is defined as:

$$k_r[Z(t)] = R_r[Z(t_1), Z(t_2), \dots, Z(t_r)]_{t_1 = t_2 = \cdots = t_r = t} \tag{6}$$

As recently recognised [22], by Eq. (5) it is possible to evaluate the cumulant of rth order independently from the evaluation of cumulants of different order than r. It is necessary to emphasise that such a property is not satisfied by the moment differential equations. Indeed, it is well known that in the rth order moment differential equation, moments of lesser order than r appear.

2.2. Multi-dimensional linear systems

In this section it will be evaluated the response cumulant vectors of multi-dimensional linear systems under multi-dimensional non-normal delta-correlated input, by using the vector state approach. To this aim, the differential equations governing the evolution of the structural response can be written as follows:

$$\dot{\boldsymbol{Z}}(t) = \boldsymbol{D}(t)\boldsymbol{Z}(t) + \boldsymbol{V}(t)\boldsymbol{W}(t) \tag{7}$$

where $\mathbf{Z}(t)$ is the n-vector listing the state variables of the structural system, while $\boldsymbol{D}(t)$ is the so-called dynamic matrix (of order $n \times n$) and $\boldsymbol{V}(t)$ is the loading matrix (of order $n \times m$). In Eq. (7) $\boldsymbol{W}(t)$ is an m-vector of independent delta-correlated processes, with the following correlation functions:

$$\boldsymbol{R}_r[\boldsymbol{W}(t_1), \boldsymbol{W}(t_2), \ldots, \boldsymbol{W}(t_r)] = \boldsymbol{q}_r(t_1)\delta(t_2 - t_1)\delta(t_3 - t_1)\cdots\delta(t_r - t_1) \tag{8a}$$

$$R_r[W_j(t_1), W_h(t_2), \ldots, W_k(t_r)] = 0\,, \quad j \neq h \quad \text{or} \quad j \neq k \quad \text{or} \quad h \neq k \tag{8b}$$

In Eq. (8a), $\boldsymbol{q}_r(t)$ is an m^r-vector containing the strength of order r of the vector process $\boldsymbol{W}(t)$ and $R_r[\cdot]$ is the vector of r-order correlations of $\boldsymbol{W}(t)$. For $r = 1, 2, 3, \ldots$ these vectors can be written as follows:

$$\boldsymbol{R}_1[\boldsymbol{W}(t_1)] = E[\boldsymbol{W}(t_1)] = \boldsymbol{q}_1(t_1) \tag{9a}$$

$$\begin{aligned} \boldsymbol{R}_2[\boldsymbol{W}(t_1), \boldsymbol{W}(t_2)] &= E[\boldsymbol{W}(t_1) \otimes \boldsymbol{W}(t_2)] - E[\boldsymbol{W}(t_1)] \otimes E[\boldsymbol{W}(t_2)] \\ &= \boldsymbol{q}_2(t_1)\delta(t_2 - t_1) \end{aligned} \tag{9b}$$

$$\begin{aligned} \boldsymbol{R}_3[\boldsymbol{W}(t_1), \boldsymbol{W}(t_2), \boldsymbol{W}(t_3)] &= E[\boldsymbol{W}(t_1) \otimes \boldsymbol{W}(t_2) \otimes \boldsymbol{W}(t_3)] - \boldsymbol{q}_1(t_1) \otimes E[\boldsymbol{W}(t_2) \otimes \boldsymbol{W}(t_3)] \\ &\quad - E[\boldsymbol{W}(t_1) \otimes \boldsymbol{q}_1(t_2) \otimes \boldsymbol{W}(t_3)] - E[\boldsymbol{W}(t_1) \otimes \boldsymbol{W}(t_2)] \otimes \boldsymbol{q}_1(t_3) \\ &\quad + 2\boldsymbol{q}_1(t_1) \otimes \boldsymbol{q}_1(t_2) \otimes \boldsymbol{q}_1(t_3) \\ &= \boldsymbol{q}_3(t_1)\delta(t_2 - t_1)\delta(t_3 - t_1) \end{aligned} \tag{9c}$$

Obviously, if $t_1 = t_2 = \cdots = t_r = t$, the relationships (9a–c) lead to the cumulant vectors of order $r = 1, 2, 3, \ldots$ respectively. In Eqs. (9), the symbol $\otimes$ means Kronecker product [23,24].

By applying the generalised Itô stochastic differential calculus, the differential equations governing the evolution of the response cumulant vectors can be written as follows:

$$\dot{\boldsymbol{k}}_r[\boldsymbol{Z}(t)] = \boldsymbol{D}_r(t)\boldsymbol{k}_r[\boldsymbol{Z}(t)] + \boldsymbol{V}^{[r]}(t)\boldsymbol{q}_r(t) \tag{10}$$

where

$$\boldsymbol{D}_r(t) = \boldsymbol{Q}_r[\boldsymbol{I}_n^{[r-1]} \otimes \boldsymbol{D}(t)] \tag{11}$$

$\boldsymbol{I}_n$ being the identity matrix of order n and the matrix $\boldsymbol{Q}_r$, of order $n^r \times n^r$, is a summation of Boolean matrices [25].

In Eq. (10), $\boldsymbol{k}_r[\boldsymbol{Z}(t)]$ is the vector of cumulants of rth order of the structural response, which is solution of a set of first order linear differential equation and can be evaluated by applying the numerical procedures recently proposed [25].

3. Approximate probability density function for one-dimensional linear systems

In order to obtain an approximate probabilistic characterisation of the response, the description of a non-Gaussian response process can be properly carried out by using an adjustable non-Gaussian probability density function, such as the so-called A-type Gram–Charlier series expansion, defined as follows [14–16]:

$$p_Z(z) = p_Z^0(z)\left[1 + \sum_{j=3}^{\infty} \frac{1}{j!} C_j[Z] H_j(z^*)\right] \tag{12}$$

where $H_j(\cdot)$ are the Hermite polynomials, $Z^* = (Z - m_Z)/\sigma_Z$ is the standardised random variable, $m_Z = k_1[Z]$

and $\sigma_Z = \sqrt{k_2[Z]}$ being the mean value and the standard deviation of the process $Z(t)$. Moreover, $p_Z^0(z)$ is the Gaussian probability density function, given as

$$p_Z^0(z) = \frac{1}{\sqrt{2\pi}\sigma_Z} \exp\left[-\frac{1}{2}\left(\frac{z - m_Z}{\sigma_Z}\right)^2\right] \tag{13}$$

In Eq. (12), the coefficients $C_j[Z]$ are the so-called Hermite moments, defined as

$$C_j[Z] = E[H_j(Z^*)] = \frac{b_j[Z]}{\sigma_Z{}^j} \tag{14}$$

related to the quasi-moments $b_j[Z]$ of the random process $Z(t)$, that can be evaluated in terms of cumulants by means of the following recursive relationship:

$$b_j[Z] = k_j[Z] + \sum_{r=3}^{j-3} \frac{(j-1)!}{r!(j-r-1)!} k_{j-r}[Z] b_r[Z] , \quad (j \geqslant 3) \tag{15}$$

Unfortunately, the accuracy of the expansion (12) is not guaranteed and sometime this series exhibits negative values of probability density, which do not have physical meaning. To overcome these drawbacks, Charlier itself has proposed the so-called C-type series expansion [16,17], defined as

$$p_Z(z) = \mathcal{N} \exp\left[\sum_{j=1}^{\infty} \frac{1}{j} \gamma_j[Z] H_j(z^*)\right] \tag{16}$$

where $\mathcal{N}$ is a normalising constant. The exponential form of this series ensures that the probability density function is definite positive; moreover, only a few terms of this series permit to approximate accurately those probability densities which strongly deviate from the Gaussian condition. The expansion (16) requires the knowledge of the coefficients $\gamma_j[Z]$, that can be related to the quasi-moments $b_j[Z]$ of the response by means of approximate relationships, as shown later.

In order to evaluate the coefficients $\gamma_j[Z]$, let us consider the expression (16) truncated at Nth term. By performing an integration by part, it can be easily seen that the following equality holds:

$$\int_{-\infty}^{+\infty} H_{i-1}(z^*) \frac{\mathrm{d}p_Z(z)}{\mathrm{d}z^*} \mathrm{d}z = -\int_{-\infty}^{+\infty} \frac{\mathrm{d}H_{i-1}(z^*)}{\mathrm{d}z^*} p_Z(z) \, \mathrm{d}z \tag{17}$$

By remembering the mathematical definition of stochastic average, by substituting in the first integral the probability density (16) truncated at N-th term and by taking into account of the property of the derivatives of the Hermite polynomials, we obtain

$$\sum_{j=1}^{N} E[H_{i-1}(z^*) H_{j-1}(z^*)] \gamma_j = -(i-1) E[H_{i-2}(z^*)] \tag{18}$$

The stochastic averages appearing on the left-hand side of Eq. (18) can be evaluated in terms of Hermite moments $C_j[Z]$, by considering that

$$H_{i-1}(z^*) H_{j-1}(z^*) = \sum_{k=0}^{i+j-2} \frac{1}{k!} \Delta_{i-1,j-1,k} H_k(z^*) \tag{19}$$

where $\Delta_{p,q,r}$ is defined as

$$\Delta_{p,q,r} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} H_p(x) H_q(x) H_r(x) \exp(-x^2/2) \, \mathrm{d}x \tag{20}$$

and can be evaluated as follows:

$$\Delta_{p,q,r} = \begin{cases} \prod_{\alpha=p,q,r} \dfrac{\Gamma(\alpha+1)}{\Gamma(\beta-\alpha+1)} , & \text{if } p+q+r = \text{even} , \quad \beta \geqslant p, q, r , \\ 0 , & \text{elsewhere} \end{cases} \tag{21}$$

with $\beta=(p+q+r)/2$ and $\Gamma(\cdot)$ being the Gamma function.

Taking into account Eqs. (14) and (19), Eq. (18) becomes

$$\sum_{j=1}^{N}\left[\sum_{k=0}^{i+j-2}\frac{1}{k!}\Delta_{i-1,j-1,k}C_k[Z]\right]\gamma_j[Z]=-(i-1)C_{i-2}[Z]\quad (i=1,2,\dots,N) \tag{22}$$

Eqs. (22) represent a set of N algebraic equations, that can be written in the following matrix form:

$$\boldsymbol{X\gamma}=\boldsymbol{\chi} \tag{23}$$

where $\boldsymbol{\gamma}=[\gamma_1\gamma_2\cdots\gamma_N]^{\mathrm{T}}$ is the N-vector of unknown coefficients of the truncated C-type Gram–Charlier series expansion, while the N components of the vector $\boldsymbol{\chi}$ and the $N(N+1)/2$ elements of the symmetric matrix $\boldsymbol{X}$ are given as follows:

$$\chi_i=-(i-1)C_{i-2}[Z]\,,\quad (\chi_1=0) \tag{24}$$

$$X_{ij}=\sum_{k=0}^{i+j-2}\frac{1}{k!}\Delta_{i-1,j-1,k}C_k[Z] \tag{25}$$

The solution of the linear system (23) gives an approximation of the first N coefficients $\gamma_j[Z]$ of the series expansion equation (16) which leads to an approximate probability density function.

4. Approximate probability density function for multi-dimensional linear systems

4.1. A-type Gram–Charlier series expansion

In order to obtain an approximate joint probability density function of the response, usually the so-called A-type Gram–Charlier series expansion is adopted. For multi-dimensional systems this series can be written as follows [1,7,14–16]:

$$p_Z(z)=p_Z^0(z)\left[1+\sum_{j=3}^{\infty}\frac{1}{j!}\boldsymbol{C}_j^{\mathrm{T}}[\boldsymbol{Z}]\boldsymbol{H}_j(z^*)\right] \tag{26}$$

where $p_Z^0(z)$ is the Gaussian joint probability density function, defined as

$$p_Z^0(z)=\frac{1}{(2\pi)^{n/2}\,\mathrm{Det}(\Sigma_Z)^{1/2}}\exp\left[-\frac{1}{2}(z-\boldsymbol{m}_Z)^{\mathrm{T}}\Sigma_Z^{-1}(z-\boldsymbol{m}_Z)\right] \tag{27}$$

$\boldsymbol{m}_Z$ and Σ_Z being the mean value vector and the covariance matrix, such that:

$$\boldsymbol{m}_Z=\boldsymbol{k}_1[\boldsymbol{Z}]\,,\qquad \mathrm{Vec}(\Sigma_Z)=\boldsymbol{k}_2[\boldsymbol{Z}] \tag{28}$$

where the symbol Vec$(\cdot)$ means vectorialised form of $(\cdot)$, i.e. it is a column vector formed by all columns of the matrix $(\cdot)$ written the one below the other. Moreover, in Eq. (26), z^* is the standardised variable vector, defined as

$$z^*=\Sigma_Z^{-1/2}(z-\boldsymbol{m}_Z) \tag{29}$$

and $\boldsymbol{H}_j(z^*)$ is the multi-dimensional Hermite polynomial vector of order j (see Appendix A). Moreover, in Eq. (24), $\boldsymbol{C}_j[\boldsymbol{Z}]$ is the Hermite moment vector of order j, defined as

$$\boldsymbol{C}_j[\boldsymbol{Z}]=E[\boldsymbol{H}_j(z^*)]=\Sigma_Z^{-1/2}\boldsymbol{b}_j[\boldsymbol{Z}] \tag{30}$$

where $\boldsymbol{b}_j[\boldsymbol{Z}]$ is the quasi-moment vector of order j of the vector process $\boldsymbol{Z}$, which can be evaluated in terms of cumulants vectors by means of recursive relationships [7].

4.2. Modified A-type Gram–Charlier series expansion

The main drawback of Eq. (26) in numerical applications lies to the fact that generally the covariance matrix is a full matrix. Consequently, the state variables appearing in the exponential function equation (27) and in the multi-dimensional Hermite polynomials are coupled. To overcome this drawback, the following coordinate transformation is here adopted:

$$\boldsymbol{Z} = \boldsymbol{\Phi Y} \tag{31}$$

where the matrix $\boldsymbol{\Phi}$ is evaluated by solving the following eigenproblem:

$$\Sigma_Z \boldsymbol{\Phi} = \boldsymbol{\Phi} \Sigma_Y \tag{32}$$

with the normality condition $\boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{\Phi} = \boldsymbol{I}_n$. In Eq. (32), Σ_Y is the diagonal covariance matrix of the vector process $\boldsymbol{Y}$. This means that the second-order cross-correlations between the components of the random vector process $\boldsymbol{Y}$ vanish. By adopting the coordinate transformation (31), Eq. (7) can be rewritten as follows

$$\dot{\boldsymbol{Y}}(t) = \boldsymbol{A}(t)\boldsymbol{Y}(t) + \boldsymbol{G}(t)\boldsymbol{W}(t) \tag{33}$$

where

$$\boldsymbol{A}(t) = \boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{D}(t)\boldsymbol{\Phi}\,, \qquad \boldsymbol{G}(t) = \boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{V}(t) \tag{34}$$

It follows that the cumulant differential equations of the vector process $\boldsymbol{Y}(t)$ can be written as follows:

$$\dot{\boldsymbol{k}}_r[\boldsymbol{Y}(t)] = \boldsymbol{A}_r(t)\boldsymbol{k}_r[\boldsymbol{Y}(t)] + \boldsymbol{G}^{[r]}(t)\boldsymbol{q}_r(t) \tag{35}$$

where, analogously to relationships (11), we can write

$$\boldsymbol{A}_r(t) = \boldsymbol{Q}_r[\boldsymbol{I}_n^{[r-1]} \otimes \boldsymbol{A}(t)] \tag{36}$$

The main advantage in using the space of the vector process $\boldsymbol{Y}(t)$ instead of the original one lies in the fact that in the A-type Gram–Charlier series expansion, the Gaussian probability density function of the vector process $\boldsymbol{Y}(t)$ can be written as products of Gaussian probability density functions of uncorrelated random processes y_i, that is

$$p_{\boldsymbol{Y}}^0(\boldsymbol{y}) = \prod_{i=1}^{n} p_{Y_i}^0(y_i) \tag{37}$$

in which

$$p_{Y_i}^0(y_i) = \frac{1}{\sqrt{2\pi}\sigma_{Y_i}} \exp\left[-\frac{1}{2}(y_i^*)^2\right] \tag{38}$$

is the Gaussian probability density function of the ith component Y_i of the random vector process $\boldsymbol{Y}$, y_i^* being the ith element of the standardised vector variable $\boldsymbol{y}^*$, given as follows:

$$y_i^* = \frac{y_i - m_{Y_i}}{\sigma_{Y_i}} \tag{39}$$

m_{Y_i} and σ_{Y_i} being the ith elements of the vector $\boldsymbol{k}_1[\boldsymbol{Y}]$ and of the diagonal matrix $\Sigma_{\boldsymbol{Y}}$, respectively.

By using the notation before introduced, the A-type Gram–Charlier series expansion can be written in the following simplified form:

$$p_{\boldsymbol{Y}}(\boldsymbol{y}) = p_{\boldsymbol{Y}}^0(\boldsymbol{y})\left[1 + \sum_{j=3}^{\infty} \frac{1}{j!} \boldsymbol{H}_j^{\mathrm{T}}(\boldsymbol{y}^*)\boldsymbol{C}_j[\boldsymbol{Y}]\right] \tag{40}$$

Due to the fact that the set of variables are uncorrelated, the elements of the Hermite polynomial vector $\boldsymbol{H}_j(\boldsymbol{y}^*)$ which appears in Eq. (40) can be written as products of one-dimensional Hermite polynomials $H_h(y_k^*)$ (see Appendix A). Moreover, in Eq. (40), the Hermite moments $\boldsymbol{C}_j[\boldsymbol{Y}]$ are given as follows:

$$\boldsymbol{C}_j[\boldsymbol{Y}] = E[\boldsymbol{H}_j(\boldsymbol{y}^*)] = (\Sigma_Y^{-1/2})^{[j]}\boldsymbol{b}_j[\boldsymbol{Y}] \tag{41}$$

where Σ_Y is a diagonal matrix and $\boldsymbol{b}_j[Y]$ is the quasi-moment vector of order j, that is related to the cumulant vectors as follows:

$$\boldsymbol{b}_j[Y]=\boldsymbol{k}_j[Y]+\frac{1}{j!}\boldsymbol{P}_j\left\{\sum_{r=3}^{j-3}\frac{(j-1)!}{r!(j-r-1)!}(\boldsymbol{k}_{j-r}[Y]\otimes\hat{\boldsymbol{k}}_r[Y])\right\} \tag{42}$$

where the matrix $\boldsymbol{P}_j$, of order $(n^j\times n^j)$, is defined in Appendix A and

$$\hat{\boldsymbol{k}}_j[Y]=\boldsymbol{k}_j[Y]+\left\{\sum_{r=1}^{j-1}\frac{(j-1)!}{r!(j-r-1)!}(\boldsymbol{k}_{j-r}[Y]\otimes\hat{\boldsymbol{k}}_r[Y])\right\} \tag{43}$$

for $\hat{\boldsymbol{k}}_1[Y]=\boldsymbol{0}$ and $\hat{\boldsymbol{k}}_2[Y]=\boldsymbol{0}$.

4.3. Proposed approximation

The A-type Gram–Charlier series expansion has the great advantage that their coefficients can be related to very simple relationships to the vector cumulants of the response for both one-dimensional and multi-dimensional random processes. Moreover, the simplicity of these relationships increases if the uncorrelated space is used. However this series has the very uncomfortable drawbacks due to the bad accuracy for strong non-Gaussian processes and negative values in some regions of the probability density function.

For these reasons, in this section the C-type Gram–Charlier series expansion is adopted to approximate the multidimensional probability density function of the response. However, to use this series in solving real problems, it needs to find the explicit relationships between the cumulants, solution of the differential equations (10) or (35) and the coefficients of this series. The aim of this section is to determine these relationships for multidimensional uncorrelated random processes. In order to do this, let us write the C-type Gram–Charlier series expansion in the space of the vector process $Y(t)$ truncated at Nth term as follows:

$$p_Y(\boldsymbol{y})=\mathcal{N}\exp\left[\sum_{j=1}^{N}\frac{1}{j}\boldsymbol{H}_j^{\mathrm{T}}(\boldsymbol{y}^*)\boldsymbol{\gamma}_j[Y]\right] \tag{44}$$

where $\boldsymbol{\gamma}_j[Y]$ is the jth coefficient vector of the C-type series expansion, of order n^j.

By remembering the main properties of Kronecker algebra [25] and the derivative of the multi-dimensional Hermite polynomial vectors (see Appendix A), it can be easily shown that an integration by part leads to the following equality:

$$\int_{R_n}[\boldsymbol{H}_{i-1}(\boldsymbol{y}^*)\otimes\nabla_{\boldsymbol{y}^*}p_Y(\boldsymbol{y})]\,\mathrm{d}\boldsymbol{y}=-\int_{R_n}\mathrm{Vec}[\nabla_{\boldsymbol{y}^*}\otimes\boldsymbol{H}_{i-1}^{\mathrm{T}}(\boldsymbol{y}^*)]p_Y(\boldsymbol{y})\,\mathrm{d}\boldsymbol{y}\quad(i=1,2,\dots,N) \tag{45}$$

where the symbol $\nabla_{\boldsymbol{y}^*}$ is the differential operator defined in Appendix A.

By taking into account Eq. (44), and the differentiation rule of the multi-dimensional Hermite polynomials (see Appendix A), after some algebra the integral appearing in the left-hand side of Eq. (45) can be written as follows:

$$\begin{aligned}\int_{R_n}[\boldsymbol{H}_{i-1}(\boldsymbol{y}^*)\otimes\nabla_{\boldsymbol{y}^*}p_Y(\boldsymbol{y})]\,\mathrm{d}\boldsymbol{y}&=\int_{R_n}\left\{\boldsymbol{H}_{i-1}(\boldsymbol{y}^*)\otimes\left[\sum_{j=1}^{N}\frac{1}{j}(\nabla_{\boldsymbol{y}^*}\otimes\boldsymbol{H}_j^{\mathrm{T}}(\boldsymbol{y}^*))\boldsymbol{\gamma}_j\right]\right\}p_Y(\boldsymbol{y})\,\mathrm{d}\boldsymbol{y}\\&=\sum_{j=1}^{N}\left\{\int_{R_n}[\boldsymbol{H}_{i-1}(\boldsymbol{y}^*)\otimes\boldsymbol{H}_{j-1}^{\mathrm{T}}(\boldsymbol{y}^*)\otimes\boldsymbol{I}_n]p_Y(\boldsymbol{y})\,\mathrm{d}\boldsymbol{y}\right\}\boldsymbol{\gamma}_j\\&=\sum_{j=1}^{N}\{E[\boldsymbol{H}_{i-1}(\boldsymbol{y}^*)\boldsymbol{H}_{j-1}^{\mathrm{T}}(\boldsymbol{y}^*)]\otimes\boldsymbol{I}_n\}\boldsymbol{\gamma}_j=\sum_{j=1}^{N}\boldsymbol{X}_{ij}\boldsymbol{\gamma}_j\end{aligned} \tag{46}$$

where the matrix $\boldsymbol{X}_{ij}$ (of order $n^i\times n^j$) is given as

$$\boldsymbol{X}_{ij}=E[\boldsymbol{H}_{i-1}(\boldsymbol{y}^*)\boldsymbol{H}_{j-1}^{\mathrm{T}}(\boldsymbol{y}^*)]\otimes\boldsymbol{I}_n=\boldsymbol{\Lambda}_{i-1,j-1}\otimes\boldsymbol{I}_n \tag{47}$$

It can be easily to shown that $\boldsymbol{X}_{ji}=\boldsymbol{X}_{ij}^{\mathrm{T}}$; moreover, the matrix $\boldsymbol{\Lambda}_{i-1,j-1}$ (of order $n^{i-1}\times n^{j-1}$) can be related to the Hermite moment vectors $\boldsymbol{C}_k[Y]$ as follows:

$$\mathrm{Vec}\{\boldsymbol{A}_{i-1,j-1}\} = E[\boldsymbol{H}_{j-1}(\boldsymbol{y}^*) \otimes \boldsymbol{H}_{i-1}(\boldsymbol{y}^*)] = \sum_{k=0}^{i+j-2} \frac{1}{k!} \boldsymbol{\Delta}_{i-1,j-1,k} \boldsymbol{C}_k[\boldsymbol{Y}] \tag{48}$$

where the matrix $\boldsymbol{\Delta}_{p,q,r}$, of order $n^{p+q} \times n^r$, is given as

$$\boldsymbol{\Delta}_{p,q,r} = \frac{1}{(2\pi)^{n/2}} \int_{R_n} [\boldsymbol{H}_q(\boldsymbol{x}) \otimes \boldsymbol{H}_p(\boldsymbol{x})] \boldsymbol{H}_r^{\mathrm{T}}(\boldsymbol{x}) \exp\left(-\frac{1}{2}\boldsymbol{x}^{\mathrm{T}}\boldsymbol{x}\right) \mathrm{d}x_1\, \mathrm{d}x_2 \cdots \mathrm{d}x_n \tag{49}$$

that can be evaluated by using Eq. (21) and taking into account that an element of the Hermite polynomial vectors $\boldsymbol{H}_i(\boldsymbol{y}^*)$ is the product of one-dimensional Hermite polynomials of the type $H_j(y_r^*)H_k(y_s^*) \cdots H_l(y_t^*)$, with $j + k + \cdots + l = i$ (see Appendix A).

Moreover, the stochastic average appearing on the right-hand side of Eq. (45) can be evaluated in the following form:

$$\begin{aligned} -\int_{R_n} \mathrm{Vec}[\nabla_{\boldsymbol{y}^*} \otimes \boldsymbol{H}_{i-1}^{\mathrm{T}}(\boldsymbol{y}^*)] p_{\boldsymbol{Y}}(\boldsymbol{y})\, \mathrm{d}\boldsymbol{y} &= -\int_{R_n} \mathrm{Vec}[(\boldsymbol{H}_{i-2}^{\mathrm{T}}(\boldsymbol{y}^*) \otimes \boldsymbol{I}_n)\boldsymbol{Q}_{i-1}] p_{\boldsymbol{Y}}(\boldsymbol{y})\, \mathrm{d}\boldsymbol{y} \\ &= -\mathrm{Vec}[(\boldsymbol{C}_{i-2}^{\mathrm{T}}[\boldsymbol{Y}] \otimes \boldsymbol{I}_n)\boldsymbol{Q}_{i-1}] = \boldsymbol{\chi}_i \end{aligned} \tag{50}$$

that holds for $i = 2, 3, \ldots, N$, while $\boldsymbol{\chi}_1 = \boldsymbol{0}$. Then, Eq. (45) written for $i = 1, 2, 3, \ldots, N$ becomes

$$\sum_{j=1}^{N} \boldsymbol{X}_{ij} \boldsymbol{\gamma}_j = \boldsymbol{\chi}_i \quad (i = 1, 2, \ldots, N) \tag{51}$$

that represent a system of $m = \Sigma_{i=1}^{N} n^i$ algebraic equations, that can be written in the following matrix form:

$$\boldsymbol{X}\boldsymbol{\gamma} = \boldsymbol{\chi} \tag{52}$$

where

$$\boldsymbol{X} = \begin{vmatrix} \boldsymbol{X}_{11} & \boldsymbol{X}_{12} & \cdots & \boldsymbol{X}_{1N} \\ \boldsymbol{X}_{21} & \boldsymbol{X}_{22} & \cdots & \boldsymbol{X}_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{X}_{N1} & \boldsymbol{X}_{N2} & \cdots & \boldsymbol{X}_{NN} \end{vmatrix}, \qquad \boldsymbol{\gamma} = \begin{vmatrix} \boldsymbol{\gamma}_1 \\ \boldsymbol{\gamma}_2 \\ \vdots \\ \boldsymbol{\gamma}_N \end{vmatrix}, \qquad \boldsymbol{\chi} = \begin{vmatrix} \boldsymbol{\chi}_1 \\ \boldsymbol{\chi}_2 \\ \vdots \\ \boldsymbol{\chi}_N \end{vmatrix} \tag{53}$$

$\boldsymbol{X}$ being a symmetric matrix, of order $m \times m$.

Once the probability density function of the vector process $\boldsymbol{Y}$ is evaluated, it is possible to evaluate that of the vector process $\boldsymbol{Z}$ as follows:

$$p_{\boldsymbol{Z}}(\boldsymbol{z}) = |\boldsymbol{\Phi}^{\mathrm{T}}| p_{\boldsymbol{Y}}(\boldsymbol{\Phi}^{\mathrm{T}}\boldsymbol{z}) \tag{54}$$

It has to be emphasised that the coefficient matrix $\boldsymbol{X}$ of the linear algebraic system given by Eq. (52) is singular. This is due to the fact that, by applying Kronecker algebra, some equations and some elements of the unknown vectors $\boldsymbol{\gamma}_j$ are equal. To overcome this drawback, one can reduce the system equation (52) by eliminating both the repeated equations and repeated unknown coefficients.

Moreover, once an approximation of the unknown vector coefficients $\boldsymbol{\gamma}_j$ is obtained, the normalisation of the approximate probability density function given by Eq. (44) requires a certain caution; in fact, by inspection of this equation, it is obvious that the exponential function which appears can not respect the limit condition $\lim_{z\to\infty} p_{\boldsymbol{Z}}(\boldsymbol{z}) = 0$. In this case, an 'ad hoc' truncation of the exponential function in the range of interest allows to perform an accurate normalisation of the approximate probability density function.

5. Application

In order to validate the proposed approximation of the probability density function of the response, as an application let us consider the dynamic behaviour of a simply supported beam of length L subjected to non-normal delta-correlated Poisson pulses acting at the midspan of the beam. The pulses occur at random time t_k, which constitute a stationary Poisson process, with constant parameter $\lambda > 0$.

The vibrations of the beam are described by the following equation:

$$EIw^{IV}(x,t) + c\dot{w}(x,t) + \rho A\ddot{w}(x,t) = \delta(x - L/2)\sum_{k=1}^{N(t)} P_k\delta(t - t_k) \tag{55}$$

where $w(x, t)$ denotes the transversal displacements of the beam, EI denotes the flexural rigidity, c denotes the damping coefficient, ρA denotes the mass density of unit length, $\delta(\cdot)$ is the Dirac delta function, roman numerals denote differentiation with respect to the spatial coordinate x, while dots denote differentiation with respect to time t. The amplitudes P_k are a family of identically distributed random variables, which are mutually independent and independent of the time instants t_k. Finally, $N(t)$ is a counting stationary Poisson process.

By using the normal mode approach, the deflection $w(x, t)$ can be expanded in the following form:

$$w(x,t) = \sum_{j=1}^{\infty} \psi_j(x)q_j(t) \tag{56}$$

where $\psi_j(x) = \sin(j\pi x/L)$ are the normal modes of free vibration for a simply supported beam, while $q_j(t)$ are the modal responses or generalised coordinates. By performing the usual coordinate transformation, we obtain the following differential equation governing the jth modal response:

$$\ddot{q}_j(t) + 2\zeta_j\omega_j\dot{q}_j(t) + \omega_j^2 q_j(t) = \frac{2\alpha_j}{M}\sum_{k=1}^{N(t)} P_k\delta(t - t_k) \tag{57}$$

where $\omega_j = (j\pi/L)^2\sqrt{EI/(\rho A)}$ is the jth radial natural frequency of the undamped structure, $\zeta_j = c/(2\rho A\omega_j)$ is the modal damping coefficient and $M = \rho AL$ is the mass of the beam. Moreover, the coefficient α_j vanishes for j even, while $\alpha_j = 1$ for $j = 1, 5, 9, \ldots$ and $\alpha_j = -1$ for $j = 3, 7, 11, \ldots$.

By using the vector state approach, Eq. (57), written for $j = 1, 3, 5$ can be expressed in the matrix form as Eq. (7). Then, the proposed procedure to approximate the probability density function of the response has been performed.

The parameter selected for the structural system are: first natural radian frequency $\omega_1 = 1$ rad/s, damping ratio $\zeta_1 = 0.25$, mass of the beam $M = 2$ kg. In order to stress the non-Gaussianity of the response process, the load parameters have been selected as follows: the expected rate of occurrences of impulsive forces $\lambda =$

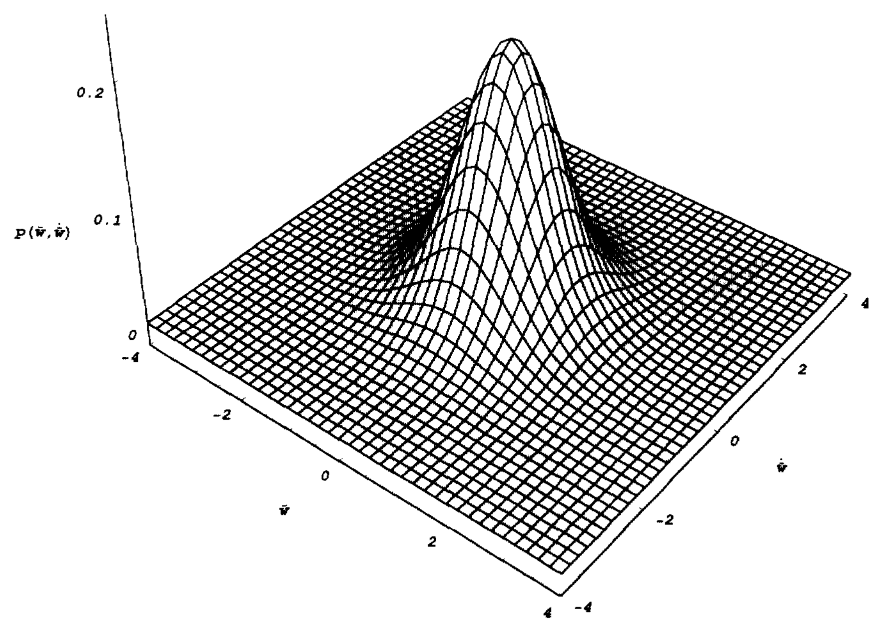

Fig. 1. Stationary joint probability density function $p(\bar{w}, \dot{\bar{w}})$ of the displacement $\bar{w}$ and the velocity $\dot{\bar{w}}$ of the midspan: C-type Gram–Charlier series expansion approximation, with $N = 6$.

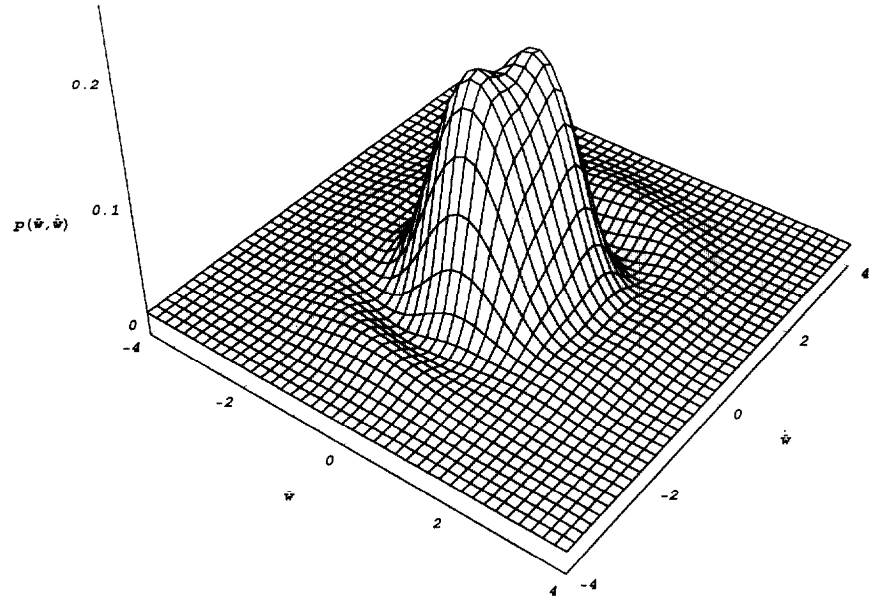

Fig. 2. Stationary joint probability density function $p(\bar{w}, \dot{\bar{w}})$ of the displacement $\bar{w}$ and the velocity $\dot{\bar{w}}$ of the midspan: A-type Gram–Charlier series expansion approximation, with $N = 10$.

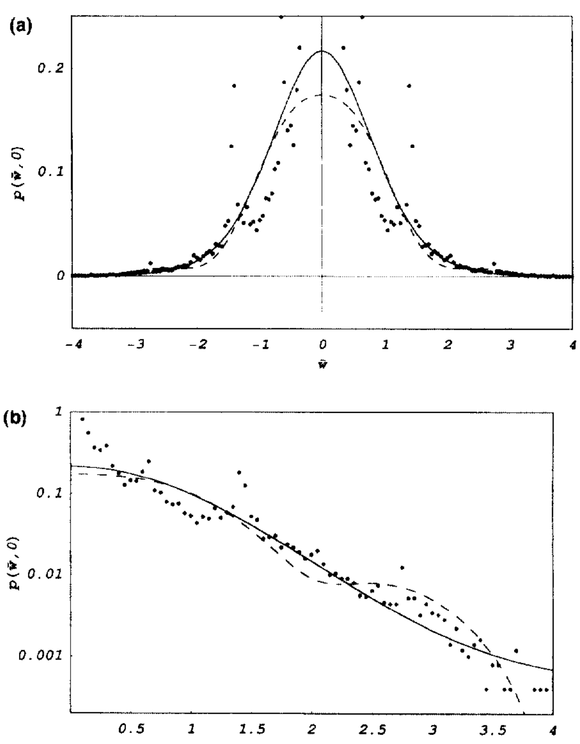

Fig. 3. Section of the stationary joint probability density function $p(\bar{w}, 0)$ of the displacement $\bar{w}$ and the velocity $\dot{\bar{w}}$ of the midspan: comparison between the C-type Gram–Charlier series expansion approximation, with $N = 6$ (solid line), the A-type Gram–Charlier series expansion approximation, with $N = 10$ (dashed line), and the Monte Carlo simulation (dots); (a) decimal scale; (b) logarithmic scale.

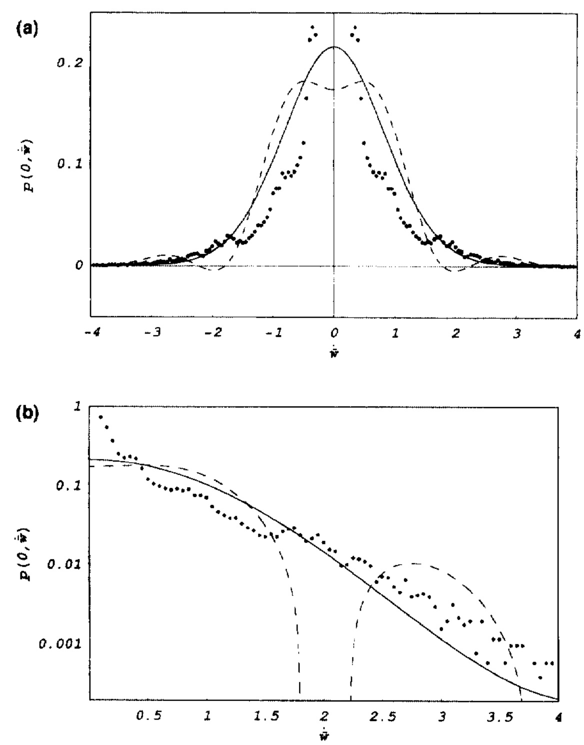

Fig. 4. Section of the stationary joint probability density function $p(0, \dot{\bar{w}})$ of the displacement $\bar{w}$ and the velocity $\dot{\bar{w}}$ of the midspan: comparison between the C-type Gram–Charlier series expansion approximation, with $N = 6$ (solid line), the A-type Gram–Charlier series expansion approximation, with $N = 10$ (dashed line), and the Monte Carlo simulation (dots); (a) decimal scale; (b) logarithmic scale.

$0.25\ \mathrm{s}^{-1}$; The intensity of the impulsive force acting on the beam as a random variable with zero mean, having the same probability to assume the values a and $-a$, with $a=\sqrt{\lambda^{-1}}$.

As an example, the probabilistic characterisation of the response has been focused on the stationary joint probability density function of the displacement $\bar{w}$ and the velocity $\dot{\bar{w}}$ of the midspan. The proposed procedure based on the C-type Gram–Charlier series expansion has been compared with the procedure that requires the use of the A-type series expansion and with Monte Carlo simulation.

The stationary probability density function $p(\bar{w}, \dot{\bar{w}})$ by using the C-type Gram–Charlier series expansion equation (44) truncated at $N=6$ has been determined; according to Eq. (49), it was necessary to evaluate the first ten cumulant vectors of the response by solving Eqs. (37), and then the first ten Hermite moment vectors, by means of Eqs. (41), (42) and (43). These last quantities have been used to approximate the function $p(\bar{w}, \dot{\bar{w}})$ by means of the A-type Gram–Charlier series expansion equation (26) truncated at $N=10$. In Figs. 1 and 2 these two approximations are plotted. To the end of comparing the results obtained through the two approaches with the Monte Carlo simulation results, the sections $p(\bar{w}, 0)$ and $p(0, \dot{\bar{w}})$ are plotted in Figs.3(a,b) and 4(a,b), for decimal and logarithmic scales. These figures show that the proposed C-type Gram–Charlier approximation leads to better results than those obtained by means of the A-type approximation. In particular the former turn out to be very close to the Monte Carlo results in correspondence of the tails; instead, the strong non-Gaussianity of the response process leads to negative values of the A-type probability density approximation. In addition, the C-type approximation gives reasonable results in the neighbourhood of the origin, where, on the contrary, the A-type approximation exhibits an anomalous behaviour.

Numerical investigations performed with a larger number of terms included in the series expansions have shown that the convergence of the C-type approximation is faster than that of the A-type one [18].

6. Conclusions

A method to approximate the probability density function of MDOF linear systems under non-normal delta-correlated input has been presented. The proposed method, based on the C-type Gram–Charlier series approximation of the probability density function, is of spread use, in particular its efficiency is remarkable when the non-Gaussianity of the response process is strong. The additional computational effort required with respect to the traditional use of the A-type Gram–Charlier approximation is negligible in comparison to the improvement of the obtained results. It is well known that a good performance in the approximation of the response probability density function is strictly required in order to investigate on further properties of the response process, such as first passage or threshold crossing.

Appendix A. Multi-dimensional Hermite polynomials

The aim of this appendix is to define the multi-dimensional Hermite polynomials introduced in this paper and to derive the main rule for the differentiation and integration of these quantities. By using extensively the Kronecker differentiation rule, the jth multi-dimensional Hermite polynomial vector can be defined as follows:

$$\boldsymbol{H}_j(\boldsymbol{x}) = (-1)^j \exp\left(\frac{1}{2}\boldsymbol{x}^{\mathrm{T}}\boldsymbol{x}\right)\nabla_x^{[j]} \exp\left(-\frac{1}{2}\boldsymbol{x}^{\mathrm{T}}\boldsymbol{x}\right) \tag{A.1}$$

where

$$\nabla_x^{\mathrm{T}} = \left[\frac{\partial}{\partial x_1}\ \frac{\partial}{\partial x_2} \cdots \frac{\partial}{\partial x_n}\right] \tag{A.2}$$

By using Kronecker differentiation law, we obtain the following recursive relationship:

$$\boldsymbol{H}_j(\boldsymbol{x}) = \boldsymbol{x} \otimes \boldsymbol{H}_{j-1}(\boldsymbol{x}) - \nabla_x \otimes \boldsymbol{H}_{j-1}(\boldsymbol{x})\,; \quad (j \geq 1) \qquad \boldsymbol{H}_0(\boldsymbol{x}) = 1 \tag{A.3}$$

It can be easily shown that:

$$\boldsymbol{H}_1(\boldsymbol{x}) = \boldsymbol{x}\,; \qquad \boldsymbol{H}_2(\boldsymbol{x}) = \boldsymbol{x}^{[2]} - \mathrm{vec}(\boldsymbol{I}_n) \tag{A.4}$$

The derivative of the transpose of the multi-dimensional Hermite polynomial vector of jth order can be given in term of the multi-dimensional Hermite polynomial vector of (j-1)th order as follows:

$$\nabla_x \otimes \boldsymbol{H}_j^{\mathrm{T}}(\boldsymbol{x}) = [\boldsymbol{H}_{j-1}^{\mathrm{T}}(\boldsymbol{x}) \otimes \boldsymbol{I}_n]\boldsymbol{Q}_j \tag{A.5}$$

where the matrix $\boldsymbol{Q}_j$, of order $n^j \times n^j$, is a summation of Boolean matrices [7,25].

The following normality condition holds:

$$\frac{1}{(2\pi)^{n/2}} \int_{R_n} \boldsymbol{H}_j(\boldsymbol{x})\boldsymbol{H}_k^{\mathrm{T}}(\boldsymbol{x}) \exp\left(-\frac{1}{2}\boldsymbol{x}^{\mathrm{T}}\boldsymbol{x}\right) \mathrm{d}x_1\, \mathrm{d}x_2 \ldots \mathrm{d}x_n = \delta_{jk}\boldsymbol{P}_j \tag{A.6}$$

where the matrix $\boldsymbol{P}_j$, of order $n^j \times n^j$, is defined as [7,25]:

$$\boldsymbol{P}_j = \boldsymbol{Q}_j(\boldsymbol{Q}_{j-1} \otimes \boldsymbol{I}_n)(\boldsymbol{Q}_{j-2} \otimes \boldsymbol{I}_n^{[2]}) \cdots (\boldsymbol{Q}_2 \otimes \boldsymbol{I}_n^{[j-2]}) \tag{A.7}$$

As an example, the first four Hermite polynomial vectors for $n = 2$ are:

$$\boldsymbol{H}_0(\boldsymbol{x}) = 1 \qquad \boldsymbol{H}_1(\boldsymbol{x}) = \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = \begin{vmatrix} H_1(x_1) \\ H_1(x_2) \end{vmatrix} \tag{A.8}$$

$$\boldsymbol{H}_2(\boldsymbol{x}) = \begin{vmatrix} x_1^2 - 1 \\ x_1 x_2 \\ x_1 x_2 \\ x_2^2 - 1 \end{vmatrix} = \begin{vmatrix} H_2(x_1) \\ H_1(x_1)H_1(x_2) \\ H_1(x_1)H_1(x_2) \\ H_2(x_2) \end{vmatrix} \qquad \boldsymbol{H}_3(\boldsymbol{x}) = \begin{vmatrix} x_1^3 - 3x_1 \\ (x_1^2 - 1)x_2 \\ (x_1^2 - 1)x_2 \\ x_1(x_2^2 - 1) \\ (x_1^2 - 1)x_2 \\ x_1(x_2^2 - 1) \\ x_1(x_2^2 - 1) \\ x_2^3 - 3x_2 \end{vmatrix} = \begin{vmatrix} H_3(x_1) \\ H_2(x_1)H_1(x_2) \\ H_2(x_1)H_1(x_2) \\ H_1(x_1)H_2(x_2) \\ H_2(x_1)H_1(x_2) \\ H_1(x_1)H_2(x_2) \\ H_1(x_1)H_2(x_2) \\ H_3(x_2) \end{vmatrix} \tag{A.9}$$

that is, an element of the multi-dimensional Hermite polynomial vector $\boldsymbol{H}_j(\boldsymbol{x})$ is a product of one-dimensional Hermite polynomials $H_r(x_s)$. It must be emphasised that this property holds if $\boldsymbol{x} = \boldsymbol{y}^* = \boldsymbol{\Sigma}_Y^{-1/2}(\boldsymbol{y} - \boldsymbol{m}_Y)$; but this does not happen if $\boldsymbol{x} = \boldsymbol{z}^* = \boldsymbol{\Sigma}^{-1/2}(\boldsymbol{z} - \boldsymbol{m}_Z)$, because the matrix $\boldsymbol{\Sigma}_Z^{-1/2}$ is full.

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